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EID. No :

Batch : 12 to 2

1. what is variable?

* A Variable is a container used to store data and value

1) var -:

value can be changed and it has function scope

2) let -:

value can be changed and it has block scope

3) const -:

value cannot be reassigned and has block scope

Example -:

```
var age = 20;
```

```
age = 21;
```

```
let mark = 50;
```

```
mark = 60;
```

```
const pi = 3.14;
```

```
// pi = 4 x
```

2. Java Script Data types -:

* Data type means what kind of value we are storing

1) String → text

```
let name = "senthil";
```

2) Number → Number

```
let age = 20;
```

3) Boolean → true or false

```
let isstudant = true;
```

4. undefined $\rightarrow$ value is not assigned
let x;

5. null $\rightarrow$ intentionally empty value
let y = null;

3. what is the different between;

1) == is called loose equality

it check the values and can convert the type

eg: $5 == "5"$

2) === is called strict equality

it checks values & data type

eg: $5 === "5"$

op/ false

h. explain these operators with examples

i) arithmetic operators

* Arithmetic operators used for mathematical calculation
*, +, -, *, /, % These are operator symbols

Example:-

let a = 10;

let b = 5;

console.log(a+b);

console.log(a-b);

op/ 15
5

ii) Assignment operators:-

* Assignment operators used to assign / values update

*, =, +=, -=, *=, /=, %=

Example:-

let x = 10;

x += 5;

op/ 5

Comparison operators:

* comparison operators used to compare various.

* Symbols : $>$, $<$, $>=$, $<=$, $==$, $===$, $!=$,
Example:

$10 > 5$

O/P:-
True

iv) logical operators:-

used to compare condition

Symbol && , || , !

• && $\rightarrow$ AND

• || $\rightarrow$ OR

! $\rightarrow$ NOT

Example:

$age \geq 18 \& \& age < 30$

O/P:-

Both condition true

5. What is the difference between
Array:-

Array $\rightarrow$ store multiple values in variable

Eg:- $\rightarrow$ let fruits = ["Apple", "mango", "orange"];

Object:-

Object store = data using key-value pairs

Eg $\rightarrow$ let student = {
name: "senhail",
age: 20
};

Array of Object:-

multiple object stored inside an array

Eg:- let student = [
{ name: "mahu", age: 20 },
{ name: "senhail", age: 21 },
];

6. What is function?

A function is a reusable block of code

That performs a specific task.

Eg:-

```
function add(a,b) {  
  return a+b;  
}
```

Parameter -

The variable written inside the function definition.

Argument -

The actual values passed when calling the function.

Return -

Send a result back from the function.

7. Even or Odd -

```
let number = 28;
```

```
if (number % 2 === 0) {
```

```
  console.log("Even");
```

```
}
```

```
else {
```

```
  console.log("odd");
```

```
}
```

O/P - 28 is odd

8. using for loop -

```
for (let i = 1; i <= 10; i++) {
```

```
  if (i % 2 === 0) {
```

```
    console.log(i);
```

```
  }
```

O/P - 2, 4, 6, 8, 10

9. factorial -

```
let number = 5;
```

```
let factorial = 1;
```

```
for (let i = 1; i <= number; i++) {
```

```
}
```

```
console.log(number + "! = " + factorial);
```

O/P -

5 ! = 120

10. fibonacci series

```
let a = 0;
let b = 1;
for (let i = 0; i < 8; i++) {
  console.log(a);
  let next = a + b;
  a = b;
  b = next;
}
```

o/p:

```
0
1
1
2
3
5
8
13
```

11. Array

```
let numbers = [10, 15, 20, 25, 30, 35, 40, 45];
for (let i = 0; i < numbers.length; i++) {
  if (console.log(numbers[i]));
}
```

o/p:

```
10
20
30
40
```

12. array total

```
let marks = [80, 75, 90, 85, 70];
let total = 0;
for (let i = 0; i < marks.length; i++) {
  total = total + marks[i];
}

console.log("total = " + total);
```

o/p:

```
total = 400
```

13. object.

```
let student = {  
  name: "senhil",  
  age: 21,  
  course: "full stack",  
  mark: 85  
}
```

```
console.log("Student Name: " + student.name);  
console.log("Course: " + student.course);  
console.log("mark: " + student.mark);
```

o/p:-

```
Student : meju :senhil  
course   :full stack  
mark     : 85
```

14. Array of Objects

```
let student = [  
  { name: "Arun", mark: 80 },  
  { name: "Kumar", mark: 45 },  
  { name: "senhil", mark: 90 },  
  { name: "meju", mark: 35 },  
]
```

```
for (let i = 0; i < student.length; i++) {  
  if (student[i].mark > 50) {  
    console.log("Students [i]: name + " + student[i].name);  
  }  
}
```

o/p

```
Arun - 80  
senhil - 90
```

15. Function - Large Number

```
function findLargest(a, b) {  
  if (a > b) {  
    return a;  
  }  
  else {  
    return b;  
  }  
}
```

```
console.log(findLargest(10, 20));
```

o/p -: